

Supplementary Material for “Exact Bayesian Inference for Spike-and-Slab Priors in Takagi–Sugeno–Kang Fuzzy Systems with Approximately Calibrated Model-Averaged Prediction Intervals”

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Notation and method definitions follow the main text (Sections 2–3). All TSK methods use $R = 5$ rules from fuzzy c -means ($m = 2$) with Gaussian membership spreads equal to the per-cluster standard deviation of training points scaled by $s = 1.5$. Results are averaged over 30 independent 80/20 train–test splits with a fixed seed unless noted otherwise.

1. Full MAE Values

Table 1: MAE comparison across all methods and targets (mean $\pm$ std, 30 splits). Best per target in **bold**.

Method	Energy-Heating	Energy-Cooling	Concrete
TSK-LS	1.32 $\pm$ 0.13	1.79 $\pm$ 0.13	6.65 $\pm$ 0.36
Bayesian-TSK	1.32 $\pm$ 0.13	1.79 $\pm$ 0.13	6.64 $\pm$ 0.36
TSK-SpikeSlab-BIC	15.72 $\pm$ 2.98	18.24 $\pm$ 2.74	43.16 $\pm$ 4.00
TSK-SpikeSlab-Gibbs	1.32 $\pm$ 0.13	1.78 $\pm$ 0.13	6.62 $\pm$ 0.36
Random Forest	0.33$\pm$0.03	1.02 $\pm$ 0.08	3.61$\pm$0.24
SVR	1.80 $\pm$ 0.16	2.05 $\pm$ 0.16	7.77 $\pm$ 0.33
Gaussian Process	0.34 $\pm$ 0.03	0.92$\pm$0.06	4.01 $\pm$ 0.25

The MAE ordering mirrors the R^2 ordering of the main text: the three dense Bayesian/least-squares TSK methods are effectively tied, the BIC-based approximation collapses, and the non-interpretable references (Random Forest and Gaussian Process) attain the lowest MAE.

2. Slab Variance Sensitivity (Coefficient-Level SSVS)

Table 2 reports the coefficient-level SSVS ablation on the Energy-Cooling target (10 splits, $\pi = 0.5$, $R = 5$ rules). Varying the slab variance τ^2 from 0.1 to 100 (three orders of magnitude) spans the full sparsity–accuracy trade-off. A tight slab ($\tau^2 = 0.1$) over-shrinks the coefficients and collapses accuracy to $R^2 = -5.42$. A diffuse slab ($\tau^2 = 100$), in contrast, merely recovers the dense baseline ($R^2 = 0.92$ versus 0.94 for TSK-LS) while pruning nothing. No τ^2 setting improves on the dense baseline.

Table 2: Coefficient-level SSVS τ^2 ablation on Energy-Cooling (10 splits). Dense TSK-LS reference: $R^2 = 0.94$.

τ^2	R^2	RMSE	PICP	MPIW
0.1	-5.42	24.11	1.00	94.57
0.3	-1.57	15.04	0.96	59.42
1.0	0.88	3.24	0.93	12.55
3.0	0.91	2.90	0.90	11.03
10.0	0.91	2.85	0.89	10.76
100.0	0.92	2.62	0.91	9.88

3. Irrelevant-Feature Ablation

Table 3 reports the irrelevant-feature regime on the Energy-Cooling target (10 splits): up to 30 standard-normal noise features are appended to the $d = 8$ inputs. The dense TSK-LS degrades only modestly (from $R^2 = 0.93$ to 0.88, where the 0.93 is the 10-split dense baseline), and neither rule-level (TSK-SpikeSlab-Gibbs) nor coefficient-level (TSK-SSVS) sparsity improves upon it. Correct fuzzy c -means already provides robustness to irrelevant features, so sparsity does not improve on the dense baseline in this low-dimensional regime.

Table 3: Irrelevant-feature ablation on Energy-Cooling (R^2 ; dense TSK-LS vs rule-level Gibbs vs coefficient-level SSVS).

Appended noise features	TSK-LS	TSK-SpikeSlab-Gibbs	TSK-SSVS
0	0.932	0.932	0.884
4	0.945	0.934	0.881
12	0.878	0.878	0.876
30	0.875	0.875	0.834

4. Rule-Count Sensitivity

Table 4 sweeps the rule count R from 3 to 10 on all three targets (10 splits). Two conclusions follow. First, $R = 5$ is a reasonable default. The dense (TSK-LS) and conjugate (Bayesian-TSK) accuracy saturates by $R \approx 5-7$ on every target. The Gibbs sampler tracks the dense baseline at every R , providing a consistency check of the sampler outside its default configuration. Second, the sparsity boundary is robust to the rule count. Even at $R = 10$ (an over-parameterized model with $R(d+1) = 90$ consequent coefficients), the Gibbs sampler keeps nearly all rules active (mean 9.7/10). Rule-level sparsity thus does not activate even when the model is heavily over-parameterized. The BIC-based approximation (TSK-SpikeSlab-BIC) collapses to negative R^2 at every rule count.

Table 4: Rule-count sweep (R^2 , mean over 10 splits). “Rules” is the mean number of active rules of the Gibbs sampler (dense methods always keep R).

R	Energy-Heating					Energy-Cooling					Concrete				
	LS	Bayes	BIC	Gibbs	Rules	LS	Bayes	BIC	Gibbs	Rules	LS	Bayes	BIC	Gibbs	Rules
3	0.949	0.948	-2.17	0.948	2.6	0.918	0.917	-12.88	0.916	2.7	0.611	0.625	-5.34	0.673	3.0
4	0.983	0.979	-0.47	0.976	4.0	0.950	0.946	-1.51	0.939	4.0	0.714	0.717	-6.09	0.729	4.0
5	0.970	0.970	-1.37	0.970	4.9	0.932	0.932	-2.53	0.932	4.9	0.747	0.747	-5.82	0.747	5.0
7	0.967	0.967	-7.16	0.966	6.6	0.929	0.928	-9.71	0.927	6.6	0.750	0.748	-6.40	0.748	7.0
10	0.976	0.975	-5.79	0.973	9.7	0.939	0.939	-8.82	0.934	9.7	0.758	0.773	-2.34	0.777	9.7

5. Ablation Isolating the Failure of the BIC Baseline

Table 5 isolates which step of the BIC-plus-Laplace baseline causes its collapse on Energy-Cooling (10 splits, $R = 5$). Four configurations are compared:

1. **BIC weights + BMA** – all 2^R rule subsets are weighted by $\exp(-\text{BIC}/2)$ and averaged, so no hard threshold and no Laplace covariance are used;
2. **BIC-PIP + threshold + Laplace** – the per-rule BIC inclusion probabilities are hard-thresholded at 0.5, the selected model is fit by least squares, and the Laplace predictive covariance is used;
3. **Gibbs-PIP + threshold + Laplace** – the same prediction as (2), but the inclusion probabilities come from the Gibbs sampler rather than BIC;
4. **Full BIC baseline** – the as-implemented baseline of the main text, which additionally fits the selected model with ridge regularization.

Three conclusions follow. First, averaging is the decisive change. BIC weights fed through BMA (row 1) restore dense accuracy and near-nominal calibration ($R^2 = 0.95$, PICP 0.92) even though the model probabilities still come from BIC. Hard-thresholding those same probabilities to a single model (row 2) underperforms ($R^2 = 0.60$). Second, the Gibbs inclusion probabilities are reliable: hard-thresholding and Laplace prediction built on them (row 3) reach the dense baseline ($R^2 = 0.94$, PICP 0.94). Third, the as-implemented baseline (row 4) collapses to $R^2 = -2.5$, with part of the gap attributable to its ridge-regularized least-squares fit of the highly collinear TSK design matrix (condition number $\sim 10^{18}$). Gibbs sampling with BMA is the only configuration that is simultaneously accurate and calibrated.

Table 5: Ablation of the BIC-plus-Laplace failure mechanism on Energy-Cooling (mean over 10 splits, $R = 5$). Dense TSK-LS reference: $R^2 = 0.94$.

Configuration	R^2	PICP	MPIW
BIC weights + BMA (no threshold)	0.95	0.92	7.8
BIC-PIP + threshold + Laplace (LS)	0.60	0.93	26.1
Gibbs-PIP + threshold + Laplace (LS)	0.94	0.94	8.4
Full BIC baseline (as implemented)	-2.53	0.00	9.9

6. MCMC Convergence Diagnostics

Table 6 reports multi-chain convergence diagnostics for the rule-level Gibbs sampler (TSK-SpikeSlab-Gibbs) on a representative split of two targets. Three chains are run with independent

seeds (42, 123, 456), each with 1,000 burn-in and 2,000 retained draws. The Gelman–Rubin statistic satisfies $\hat{R} \leq 1.001$ for the consequents (β) and the noise variance (σ^2), and the effective sample size exceeds 1,700 for σ^2 , indicating well-mixed β and σ^2 chains. The inclusion-indicator chains are (near-)constant on these benchmarks—posterior inclusion probability saturates near 1.0 for every rule—so their effective sample size equals the chain length by construction rather than from active exploration. On Concrete the γ chain is exactly constant. Diagnostics are reported for the rule-level sampler. On these benchmarks the γ chains of the coefficient-level SSVS also saturate near full activation. We expect qualitatively similar mixing but do not claim formal convergence verification for those runs.

Table 6: MCMC convergence diagnostics for TSK-SpikeSlab-Gibbs (three chains, 1,000 burn-in, 2,000 retained draws). $\hat{R}$ is the Gelman–Rubin statistic; ESS is the effective sample size.

Target	$\hat{R}(\gamma)$ (max)	$\hat{R}(\beta)$ (median / max)	$\hat{R}(\sigma^2)$	ESS(σ^2)
Energy-Cooling	1.000	1.000 / 1.001	1.000	1,725
Concrete	constant (= 1.0)	1.000 / 1.001	1.000	1,744

7. High-Dimensional Sparsity Boundary

Table 7 tests the sparsity boundary on a real high-dimensional regression benchmark (Superconductivity, UCI id 464; subsample of $n = 3000$ rows with $d = 81$ features, 30 splits, $R = 5$ rules). The pattern of the low-dimensional benchmarks reproduces almost exactly: rule-level sparsity (TSK-SpikeSlab-Gibbs) matches the dense baselines in R^2 (0.794 versus 0.792–0.795) while keeping calibration at the nominal level (PICP 0.940). Coefficient-level sparsity (TSK-SSVS) over-shrinks across the full slab-variance grid. Table 7 reports the two settings that bracket the low-dimensional sweep of Table 2 ($\tau^2 = 1$, collapsing R^2 to 0.674; $\tau^2 = 10$, recovering only to 0.763, still below dense). A 10-split sweep of the complete grid confirms the pattern. R^2 rises monotonically from 0.36 at $\tau^2 = 0.1$ to 0.79 at $\tau^2 = 100$. No setting improves on the dense baseline (0.792), and the most diffuse slab ($\tau^2 = 100$) merely recovers it (0.794). The sparsity boundary therefore is not a low-dimensional artifact and does not depend on a particular slab-variance choice. Whether sparsity is beneficial under different antecedent structures or much larger samples remains open. On this high-dimensional target, the rule-level Gibbs sampler also maintains near-nominal calibration (PICP 0.940) where the dense conjugate baseline slightly undercovers (PICP 0.921), consistent with the calibration contribution of the main text.

Table 7: High-dimensional sparsity boundary on Superconductivity (mean $\pm$ std over 30 splits; subsample $n = 3000$, $d = 81$, $R = 5$). PICP is the 95% interval coverage; MPIW is the mean prediction-interval width.

Method	RMSE	R^2	PICP	MPIW
TSK-LS (dense)	15.46 $\pm$ 0.82	+0.792 $\pm$ 0.024	—	—
Bayesian-TSK (dense conjugate)	15.33 $\pm$ 0.77	+0.795 $\pm$ 0.023	0.921	55.6
TSK-SpikeSlab-Gibbs	15.37 $\pm$ 0.70	+0.794 $\pm$ 0.021	0.940	59.7
TSK-SSVS ($\tau^2 = 1$)	19.35 $\pm$ 0.56	+0.674 $\pm$ 0.018	0.940	75.1
TSK-SSVS ($\tau^2 = 10$)	16.48 $\pm$ 0.55	+0.763 $\pm$ 0.018	0.935	65.0

8. Mean Prediction-Interval Width

Table 8 reports the mean prediction-interval width (MPIW) for the three probabilistic methods across all targets (30 splits). The compact widths support the calibration claim of the main text (Section 5.3, $\text{MPIW} \approx 6.7\text{--}32.9$).

Table 8: MPIW for the probabilistic methods (mean over 30 splits).

Method	Energy-Heating	Energy-Cooling	Concrete
Bayesian-TSK	6.61	9.20	32.04
TSK-SpikeSlab-Gibbs	6.69	9.40	32.87
Gaussian Process	1.75	5.18	20.46

9. Synthetic Validation of the Inclusion Full Conditional

The fixed-activation correctness check of the main text (Section 3.3) exercises the β and σ^2 updates but not the inclusion full conditional of the main text (the γ_j update of Section 3.4), because the indicators are fixed there. On the real benchmarks the indicators also saturate near full activation. We therefore validate the γ_j update on a synthetic problem with *known sparse* ground truth, where the exact posterior over rule configurations is computable.

Setup. We generate $n = 600$ samples from a synthetic TSK system with $R = 5$ rules and $d = 2$ features, in which only rules 0 and 2 have nonzero consequents (ground truth active set $\{0, 2\}$), with additive Gaussian noise of variance $\sigma^2 = 1$. The fuzzy partition is built by fuzzy c -means as in the main text. We then (i) run the rule-level block-Gibbs sampler ($\pi = 0.5$, $\tau^2 = 10$, $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$, 2,000 burn-in, 10,000 retained draws) and (ii) compute the *exact* posterior inclusion probabilities by enumerating all $2^5 = 32$ rule configurations. Each configuration’s marginal likelihood is evaluated in closed form via the Woodbury identities, and σ^2 is integrated out over its inverse-gamma prior.

Result. Table 9 reports the sampler’s inclusion probabilities against the exact values. The Gibbs sampler recovers the known ground truth. Rules 0 and 2 are active with probability ≈ 1.0 , and the inactive rules are pruned (inclusion probability ≈ 0.004), matching the exact posterior to a maximum absolute difference of 8×10^{-4} . The BMA predictive distribution also agrees with the exact model-averaged predictive (maximum absolute mean difference 0.006; median relative predictive-variance difference 2.5%). This confirms the arithmetic of the inclusion full conditional and the BMA construction against exact computation, complementing the fixed-activation check of the main text.

Table 9: Synthetic validation of the γ_j update of the main text: posterior inclusion probabilities, Gibbs sampler vs exact model averaging over all 2^R configurations ($n = 600$, $d = 2$, $R = 5$; ground-truth active rules $\{0, 2\}$).

Rule	0	1	2	3	4
Gibbs	1.0000	0.0044	1.0000	0.0017	0.0034
Exact	1.0000	0.0044	1.0000	0.0019	0.0026
Gibbs–Exact	0.0000	0.0000	0.0000	0.0002	0.0008